Name: Answer	Index No.:	CT Group: 10

PIONEER JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION
HIGHER 1

CHEMISTRY 8872/02

Paper 2 **19 September 2011**

2 hours

Candidates answer Section A on the Question Paper

Additional Materials: Data Booklet
Answer paper
Graph Paper

READ THESE INSTRUCTONS FIRST

Write your name, CT group and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams, graphs or rough working.

Section A
Answer **all** questions.

Section B
Answer any **two** questions on separate answer paper.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

FOR EXAMINER'S USE			
Section A		**Section B**	
1	/ 15	5	/ 20
2	/ 10	6	/ 20
3	/ 8	7	/ 20
4	/ 7	Penalty	s.f. / units
		TOTAL	**/ 80**

Section A (40 marks)

Answer all questions in the spaces provided.

1 (a) Nitric acid is a highly corrosive liquid and a strong oxidising agent. It can oxidise $Fe^{2+}(aq)$ to $Fe^{3+}(aq)$.

In an experiment, 25.0 cm^3 of 0.600 mol dm^{-3} $Fe^{2+}(aq)$ required 25.0 cm^3 of 0.200 mol dm^{-3} HNO_3 for complete reaction.

(i) Complete the table below which refers to possible reduction products of nitric acid.

Formula of product	Oxidation number of nitrogen
NO_2	+4
NO	+2
N_2O	+1
NH_4^+	-3

(ii) How many moles of Fe^{2+} react with one mole of HNO_3?

Amount of Fe^{2+} reacted = 0.0150 mol

Amount of HNO_3 used = 5.00×10^{-3} mol

Amount of Fe^{2+} that reacts with one mole of HNO_3 = 3 mol

(iii) What change in oxidation number does the nitrogen in HNO_3 undergo?

$Fe^{2+} \rightarrow Fe^{3+} + e$

Change in oxidation number of nitrogen in HNO_3 = -3

(iv) Identify the nitrogen-containing product of this reaction from the table in **(i)**.

Final oxidation state of N in product = 5 – 3 = +2
The product of this reaction is NO.

(iv) Construct an equation for the reaction of HNO_3 with Fe^{2+}.

$HNO_3 + 3e + 3H^+ \rightarrow NO + 2H_2O$

$3Fe^{2+} + HNO_3 + 3H^+ \rightarrow 3Fe^{3+} + NO + 2H_2O$

[6]

(b) In the gaseous state, NO_2, which is used in the preparation of nitric acid, can dimerise as given in the equation.

$$2NO_2(g) \rightleftharpoons N_2O_4(g)$$

(i) Write an expression for the equilibrium constant, K_c, for the dimerisation of NO_2.

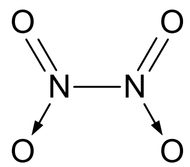
$$K_c = \frac{[N_2O_4]}{[NO_2]^2}$$

(ii) When 1 mol of NO_2 is placed in a 2 dm^3 vessel at room temperature, it is found that the equilibrium mixture contains 0.35 mol of N_2O_4. Calculate a value of K_c for the reaction, giving its units.

	$2NO_2(g)$	$\rightleftharpoons$	$N_2O_4(g)$
Initial amount / mol	1		0
Change in amount / mol	-0.70		+0.35
Equilibrium amount / mol	0.30		0.35

$$K_c = \frac{\left(\frac{0.35}{2}\right)}{\left(\frac{0.30}{2}\right)^2} = 7.78 \text{ mol}^{-1} \text{ dm}^3$$

(iii) The NO_2 molecule has an unpaired electron on the nitrogen atom but the N_2O_4 molecule does not. Using the information, suggest a structural formula for N_2O_4. Hence, explain whether the dimerisation of NO_2 is endothermic or exothermic.



Dimerisation of NO_2 involves the formation of N-N bond. Since <u>bond formation is exothermic</u>, dimerisation of NO_2 is <u>exothermic</u>.

(iv) Explain, with reasons, whether the dimerisation above is favoured by
(I) high or low pressure

Dimerisation results in the formation of less number of moles of gases. According to Le Chatelier's principle, high pressure favours the side of the reaction with less number of moles of gases. Thus, dimerisation is favoured by <u>high pressure</u>.

(II) high or low temperature

Dimerisation is an exothermic reaction. According to Le Chatelier's principle, low temperature favours an exothermic reaction. Thus, dimerisation is favoured by low temperature.

[9]

[Total: 15]

2 Magnesium is a member of the alkaline earth metals. It occurs in nature, only in the combined state, as *dolomite*, $CaCO_3MgCO_3$, *kieserite*, $MgSO_4$, *carnallite*, $KCl.MgCl_2.6H_2O$, *Epsom salt*, $MgSO_4.7H_2O$ and many silicates including talc and asbestos.

Magnesium compounds are also present in sea water. Being the second most abundant cation in sea water, it is estimated that magnesium is present in concentrations as high as 1290 parts per million (ppm) and that makes it an attractive commercial source of magnesium. [1 ppm of magnesium = 1 mg of magnesium in 1 dm^3 of solution]

About 70% of the Earth's surface is covered by water. The oceans of the planet have an area of about 360 x 10^6 km^2 with an average depth of about 3.7 km.

As of 2005, China has taken over United States as the dominant supplier of magnesium, pegged at 60% of the world's market share. Unlike the typical use of electrolysis to extract magnesium, China makes use of silicothermic reduction, known as Pidgeon Process. Calcium hydroxide is added to sea water to precipitate magnesium hydroxide. The magnesium hydroxide is then calcined to give magnesium oxide by removal of water. The magnesium oxide and silicon is then heated strongly according to the equation below.

$$Si(s) + 2MgO(s) \rightleftharpoons SiO_2(s) + 2Mg(g)$$

This reaction is not thermodynamically favourable. However, applying Le Chatelier's principle, the position of equilibrium can be shifted to the right by continuous supply of heat and distilling out the magnesium vapour formed. The magnesium extracted through this process is known to have very high purity.

Substance	Melting Point / ºC	Boiling Point / ºC
Mg	650	1091
MgO	3125	3873
Si	1414	3265
SiO_2	1700	2230

One of the main applications of magnesium is in the manufacturing of alloys such as duralumin and magnalium. These alloys are used for the construction of aircrafts and racing cars. Magnesium is also used in flares and fire-works.

(a) Use the data above to estimate the volume of the Earth's oceans in dm^3. [2]

Volume of Earth's ocean = $360 \times 10^6 \times 3.7 = 1.332 \times 10^9$ km^3
1 m = 10 dm

1 km = 1000 m = 10000 dm
1 km^3 = 10000^3 dm^3

Volume of Earth's ocean = $1.332 \times 10^9 \times 1 \times 10^{12} = 1.332 \times 10^{21}$ dm^3

(b) Calculate the concentration, in g dm^{-3}, of magnesium in seawater. [1]

Concentration of magnesium in seawater = 1290 / 1000 = 1.29 g dm^{-3}

(c) Estimate the total mass of magnesium in the ocean. [1]

Total mass of magnesium in ocean = $1.332 \times 10^{21} \times 1.29 = 1.72 \times 10^{21}$ g

(d) Account for the high melting and boiling point of magnesium in terms of its structure and bonding. [2]

Magnesium has a giant lattice of Mg^{2+} ions in a sea of delocalised, mobile electrons. As strong electrostatic forces of attraction (metallic bonds) exist between the electrons and Mg^{2+} ions, large amount of energy is required to overcome them, accounting for its high melting and boiling point.

(e) **(i)** Suggest a minimum and maximum temperature for the Pidgeon process.

Min: 1091ºC, Max: 1413ºC

(ii) Explain why magnesium obtained through this process has very high purity.

The other reactants and product (Si, MgO and SiO_2) are in the solid state, hence they will not contaminate Mg which is distilled out.

[2]

(f) Suggest a property of magnesium of its use in

(i) making of alloys for aircrafts and racing cars

Magnesium is light or magnesium adds strength to the alloys

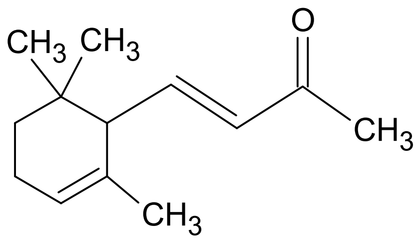
(ii) flares and fireworks

Magnesium burns with a bright white light.

[2]

[Total: 10]

3 Ionones are a significant contributor to the aroma of roses. They are found in a variety of essential oils and are important chemicals in perfumery. One such ionones is the α-ionone as shown below.



(a) Identify the functional groups present in α-ionone. [1]

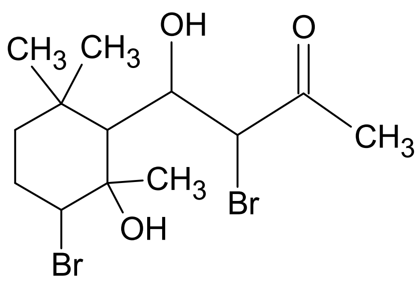
alkene and ketone

(b) State what you would see and give the structural formula of the organic products when α-ionone reacts with

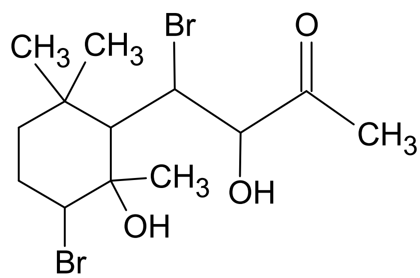
(i) aqueous bromine

Observation: orange aqueous bromine decolourises.

Organic product:



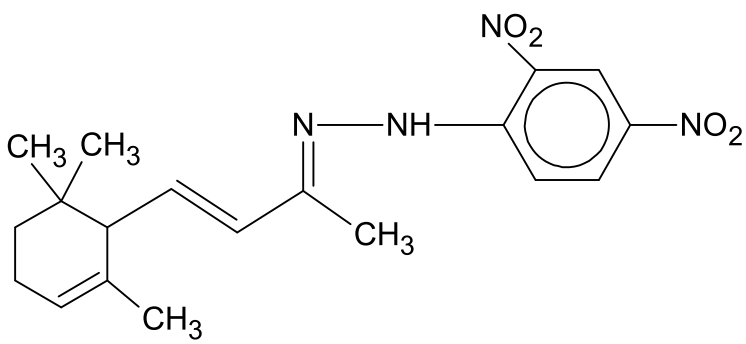
or



(ii) 2,4-dinitrophenylhydrazine

Observation: orange precipitate observed**.**

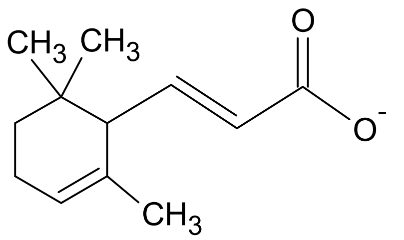
Organic product:

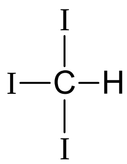


(iii) alkaline aqueous iodine

Observation: yellow precipitate observed.

Organic products:

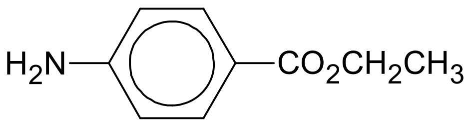




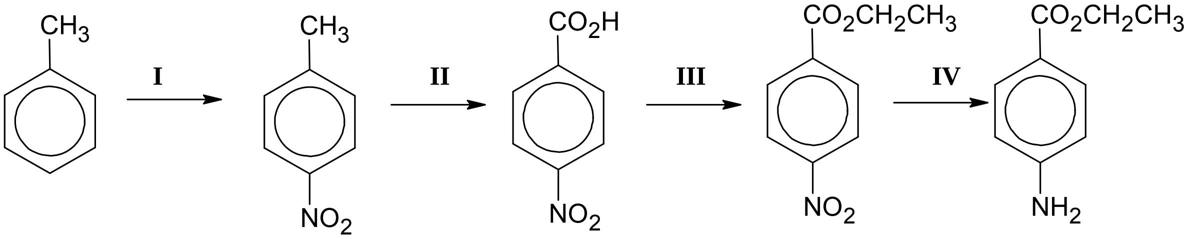
[7]

[Total: 8]

4 Benzocaine is a local anaesthetic used in sunburn ointments. Its formula is given below.



(a) Benzocaine can be made by the following reaction scheme, starting from methylbenzene.



(i) Give the reagents and conditions used for each of the stages **I** to **III**.

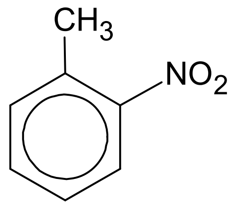
Stage **I**: concentrated nitric acid, concentrated sulfuric acid, 30ºC
Stage **II**: $KMnO_4$, H_2SO_4(aq), heat
Stage **III**: CH_3CH_2OH, concentrated sulfuric acid, heat

(ii) State the type of reaction undergone in stage **IV**.

Reduction

[4]

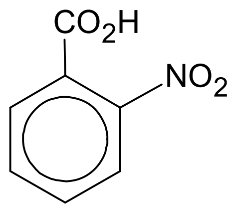
(b) (i) Besides 4-nitromethylbenzene, another isomer, 2-nitromethylbenzene is formed in stage **I**.



Identify the type of isomerism involved.

Positional isomerism

(ii) When 2-nitromethylbenzene is subjected to the same set of reagents and conditions in stage **II**, 2-nitrolbenzoic acid is obtained.



Which of the two carboxylic acids, 2-nitrobenzoic acid or 4-nitrobenzoic acid, would you expect to have a lower melting point? Explain your answer.

2-nitrobenzoic acid would have a <u>lower melting point</u>. Unlike 4-nitrobenzoic acid, the carboxylic acid group and nitro group of 2-nitrobenzoic acid are close to each other. Thus, they are able to form <u>intramolecular hydrogen bonds</u>. This <u>reduces the extent of intermolecular hydrogen bonds</u> between 2-nitrobenzoic acid molecules. <u>Less energy</u> is required to overcome the intermolecular hydrogen bonds, accounting for it lower melting point.

[3]

[Total: 7]

Section B (40 marks)

Answer **two** questions from this section on separate answer paper.

5 This question is about organic acids.

(a) **(i)** Carboxylic acid, **A**, is the main component of vinegar with a distinctive sour taste and pungent smell. On combustion, 0.00250 mol of **A** reacted exactly with 120 cm^3 of oxygen to produce 120 cm^3 of carbon dioxide and 0.090 g of water at room temperature and pressure. Determine the molecular formula and hence identify **A**.

Amount of O_2 used = Amount of CO_2 produced
= 120 / 24000 = 0.00500 mol **[½]**

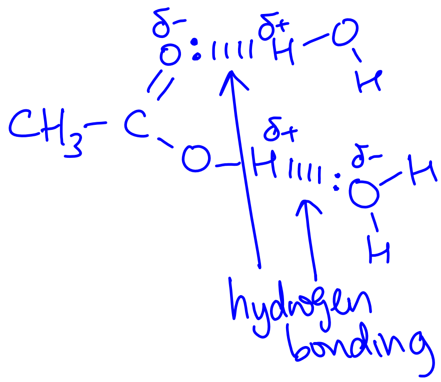
Amount of H_2O produced = 0.090 / 18.0 = 0.00500 mol **[½]**

	A +	O_2 →	CO_2 +	H_2O
	0.00250	0.00500	0.00500	0.0500
Mol ratio **[½]**	1	2	2	2

Number of C present in **A** = 2
Number of H present in **A** = 4
Number of O present in **A** = 4 – 2
= 2

Molecular formula of **A** is $C_2H_4O_2$ and **A** is ethanoic acid.

(ii) Explain, with the aid of a diagram, why **A** is soluble in water.



A is soluble in water because its molecules are able to form hydrogen bonding with water molecules.

[5]

(b) Lactic acid, $CH_3CH(OH)COOH$, also known as milk acid, plays a role in various biochemical processes

(i) Show, by means of calculations, that lactic acid is a weak acid.

$[H_3O^+] = 10^{-2.43} = 0.003715$ mol dm^{-3}

Since $[H_3O^+]$ is much smaller than 0.100 mol dm^{-3}, lactic acid undergoes partial dissociation in water to give H_3O^+, hence it is a weak acid.

(ii) Write an expression for the acid dissociation constant, K_a, of lactic acid. Hence calculate a value for K_a of lactic acid.

$$K_a = \frac{[CH_3CH(OH)COO^-][H_3O^+]}{[CH_3CH(OH)COOH]}$$
$$= \frac{0.003715^{\ 2}}{0.10-0.003715}$$
$$= 1.43 \times 10^{-4} \text{ mol dm}^{-3}$$

(iii) When sodium lactate is added to lactic acid, a buffer solution is obtained. Explain, with the aid of equations, how the mixture can act as a buffer on the addition of a small amount of acid and alkali.

When a small amount of acid is added, $CH_3CH(OH)COO^-$ removes the H^+.
$CH_3CH(OH)COO^- + H^+ \rightarrow CH_3CH(OH)COOH$

When a small amount of alkali is added, the lactic acid helps to remove the OH^-.
$CH_3CH(OH)COOH + OH^- \rightarrow CH_3CH(OH)COO^- + H_2O$

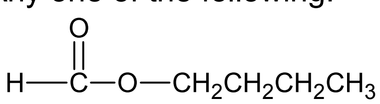
Hence pH remains relatively constant.

[6]

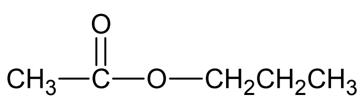
(c) Valeric acid or pentanoic acid, $CH_3CH_2CH_2CH_2COOH$, is found naturally in the perennial flowering plant, *valerian*, and has an unpleasant odour.

(i) Give a functional group isomer of pentanoic acid.

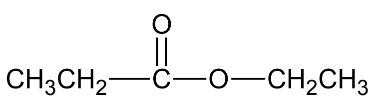
Any one of the following:

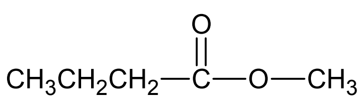


or branched isomers



or branched isomers





(ii) Pentanoic acid may be formed by heating **B** with acidified $K_2Cr_2O_7$. Suggest an identity of **B**, state the observation and give a balanced equation for this reaction.

B is $CH_3CH_2CH_2CH_2CH_2OH$ or $CH_3CH_2CH_2CH_2CHO$

Orange $K_2Cr_2O_7$ turns green.

$CH_3CH_2CH_2CH_2CH_2OH + 2[O] \rightarrow CH_3CH_2CH_2CH_2COOH + H_2O$ or
$CH_3CH_2CH_2CH_2CHO + [O] \rightarrow CH_3CH_2CH_2CH_2COOH$

(iii) Give the structural formulae of the organic products formed when pentanoic acid reacts with:

(I) Na

(II) CH_3OH, concentrated H_2SO_4, heat

(I) $CH_3CH_2CH_2CH_2COO^-Na^+$

(II) $CH_3CH_2CH_2CH_2COOCH_3$

(iv) Pentanoic acid can undergo chlorination to form chlorine-containing carboxylic acids. Suggest, with an explanation, whether 2-chloropentanoic acid or 4-chloropentanoic acid has a lower pK_a value.

2-chloropentanoic acid has a lower pK_a value as it is a stronger acid.

2-chloropentanoic acid is a stronger acid than 4-chloropentanoic acid because:

- C*l* atom is electronegative / electron-withdrawing
- C*l* in 2-chloropropanoic acid is nearer to the –COOH group and the negative charge of 2-chloropentanoate ion is more dispersed than that of 4-choropentanoate ion
- 2-choropentanoate ion is more stable than 4- choropentanoate ion. Thus, 2-chloropentanoic acid and has a greater tendency to dissociate to form 2-chloropentanoic acid and H^+.

[9]

[Total: 20]

6 This question is about silicon and its compounds.

(a) **(i)** Give the full electronic configuration of silicon.

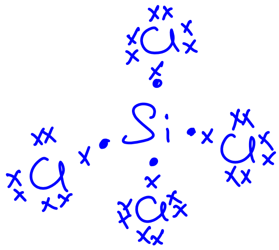
$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$

(ii) Explain why the first ionisation energy of silicon is less than that of carbon.

Si has an additional inner filled shell as compared to C. Hence the valence electron of Si is further from nucleus and less attracted by the nucleus. Thus, less energy is required to remove the valence electron of Si.

[3]

(b) **(i)** Draw the dot-and-cross diagram of $SiCl_4$. State the shape of the molecule and give its bond angle.



Shape: Tetrahedral
Bond angle: 109.5^o

(ii) $SiCl_4$ has a melting point of -70^oC, but SiO_2 melts at 1700^oC. Account for the difference in melting point in terms of structure and bonding.

SiO_2 has a <u>giant molecular structure</u> that requires a lot of energy to break the <u>strong covalent bonds</u> between <u>atoms</u> during melting.

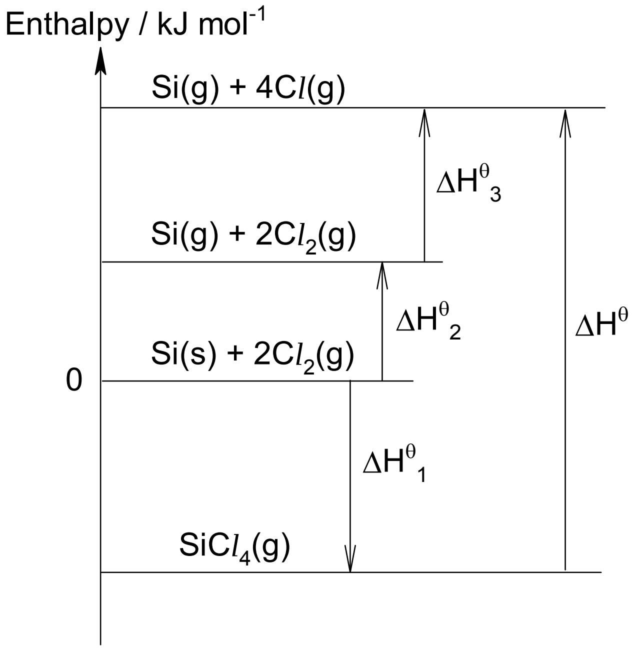
$SiCl_4$ has a <u>simple molecular structure</u> that requires little energy to break the <u>weak Van der Waals' forces</u> between <u>molecules</u> during melting.

(iii) Write an equation for the reaction of $SiCl_4$ with water and state the pH of the resulting solution.

$SiCl_4 + 2H_2O \rightarrow SiO_2 + 4HCl$
pH = 1 or 2

[6]

(c) The energy cycle below can be used to calculate the average Si-C*l* bond energy in $SiCl_4$ molecule.



(i) What is meant by the term *bond energy*?

Bond energy is the heat energy absorbed to break one mole of covalent bonds between atoms in gaseous molecules to give gaseous atoms.

(ii) Name the standard enthalpy change represented as ΔH^θ_1.

Standard enthalpy change of formation of $SiCl_4$.

(iii) Using the information provided below, calculate a value for the average Si-C*l* bond energy in $SiCl_4$ molecule.

ΔH^θ_1 = -610 kJ mol^{-1}
ΔH^θ_2 = +388 kJ mol^{-1}
ΔH^θ_3 = +488 kJ mol^{-1} [4]

Applying Hess' Law,
4 BE (Si-C*l*) = -(-610) + (+338) + 4(+122)
4 BE (Si-C*l*) = +1436
Ave BE (Si-C*l*) = +1436 / 4 = + 359 kJ mol^{-1}

(d) $SiCl_4$ reacts with CH_4 to form SiC films at 1400°C:

$$SiCl_4(g) + CH_4(g) \rightarrow SiC(s) + 4HCl(g)$$

Using the information provided below, calculate a value of the enthalpy change for the above reaction.

Enthalpy change of formation of $SiCl_4$(g) = –687 kJ mol^{-1}
Enthalpy change of formation of CH_4(g) = –74 kJ mol^{-1}

Enthalpy change of formation of SiC(s) = –65 kJ mol^{-1}
Enthalpy change of formation of $HCl(g)$ = –92 kJ mol^{-1}

[2]

$\Delta H_{reaction} = [\Delta H_f(SiC) + 4\Delta H_f(HCl)] - [\Delta H_f(SiCl_4) + \Delta H_f(CH_4)]$
$= [(-65) + 4(-92)] - [(-687) + (-74)]$
$= +328$ kJ mol^{-1}

(e) **X** is another element of the third period and it forms a chloride, $\mathbf{X}Cl_y$.

$\mathbf{X}Cl_y$ is a liquid which has a boiling point of 76°C and fumes in air. After mixing 0.010 mol of $\mathbf{X}Cl_y$ with water, the resulting solution required 100 cm^3 of 0.30 mol dm^{-3} silver nitrate for complete precipitation of the chloride ion.

(e) **(i)** Predict the type of structure and bonding in the chloride, $\mathbf{X}Cl_y$.

$\mathbf{X}Cl_y$ is a liquid with a low boiling point, suggesting that it has a simple molecular structure with weak van der Waals' intermolecular forces of attraction.

(ii) Calculate a value of y.

Amount of Ag^+ required for complete precipitation of the chloride
= 0.30 x 100/1000
= 0.0300 mol
= amount of Cl^- produced

Mole ratio of $\mathbf{X}Cl_y$: Cl^- = 0.0100 : 0.0300
= 1 : 3

Value of y = 3

(iii) Suggest, with reasons, the identity of element **X**.

Since 1 mol of $\mathbf{X}Cl_y$ reacts with water to give 3 mol of Cl^-, **X** can either be aluminium or phosphorus. Aluminium chloride is a solid whereas phosphorus trichloride is a liquid which fumes in air. Thus, **X** is phosphorus.

[5]

[Total: 20]

7 (a) Organic compound **L** is found in plants called "Hops", which is used to treat insomnia. A molecule of **L** has the molecular formula, $C_5H_{10}O$.

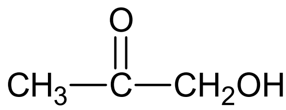
No reaction occurs between **L** and 2,4-dinitrophenylhydrazine. However, **L** produces **M**, $C_5H_{12}O$, with hydrogen with nickel catalyst.

No change is observed when **L** is warmed with acidified $K_2Cr_2O_7$.

When **L** is heated with an excess of concentrated sulfuric acid, **N**, with molecular formula C_5H_8, is produced.

N reacts with hot concentrated acidified $KMnO_4$ to produce **P**, $C_3H_4O_3$. A colourless gas, which forms a white precipitate in limewater, is liberated.

P can also be produced by the reaction of compound **Q** with acidified $KMnO_4$ and heating.

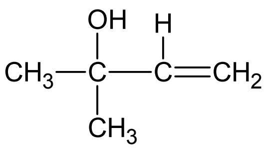


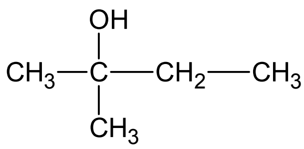
Q

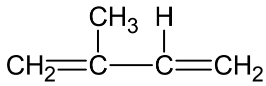
Deduce the structures of compounds **L**, **M**, **N**, and **P**. Explain the chemistry of the reactions described. [9]

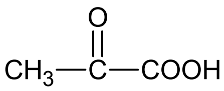
Information/ Type of reaction	Deductions: functional group
L does not undergo condensation with 2,4-dinitrophenylhydrazine.	L has is not a ketone or aldehyde.
L produces M, $C_5H_{12}O$, when it undergoes reduction with hydrogen with nickel catalyst	L is unsaturated. Additional of 2 H atoms indicate presence of a C=C bonds .
L does not undergo oxidation with acidified potassium dichromate(VI)	L contains a 3^o alcohol group.
L undergoes elimination when heated with an excess of concentrated sulfuric acid to P.	N contains an additional C=C bond upon L undergoing elimination.
N undergoes oxidation with hot concentrated acidified $KMnO_4$ to produce P, $C_3H_4O_3$. CO_2 gas which turns limewater cloudy is liberated.	Presence of 2 terminal C=C bond due to 2 moles of CO_2 produced during oxidative cleavage. (loss of 2 C from N to P.
P can also be produced by oxidation of **Q** when heated with acidified $KMnO_4$.	-

Structures:

L: 

M: 

N: 

P: 

(b) Compound **R**, CH_3CH_2COOH, is an isomer of compound **Q** from **(a)**.

An excess of small pieces of sodium carbonate was allowed to react with 100 cm^3 of 0.100 mol dm^{-3} of **R** at room temperature and pressure.

The volume of carbon dioxide gas liberated was measured at regular intervals and the results are tabulated below.

Time / s	5	10	15	20	25	30	35	45
Volume of CO_2 / cm^3	34	60	78	92	104	112	117	120

(i) Show, by means of calculations, that the maximum volume of carbon dioxide liberated was 120 cm^3.

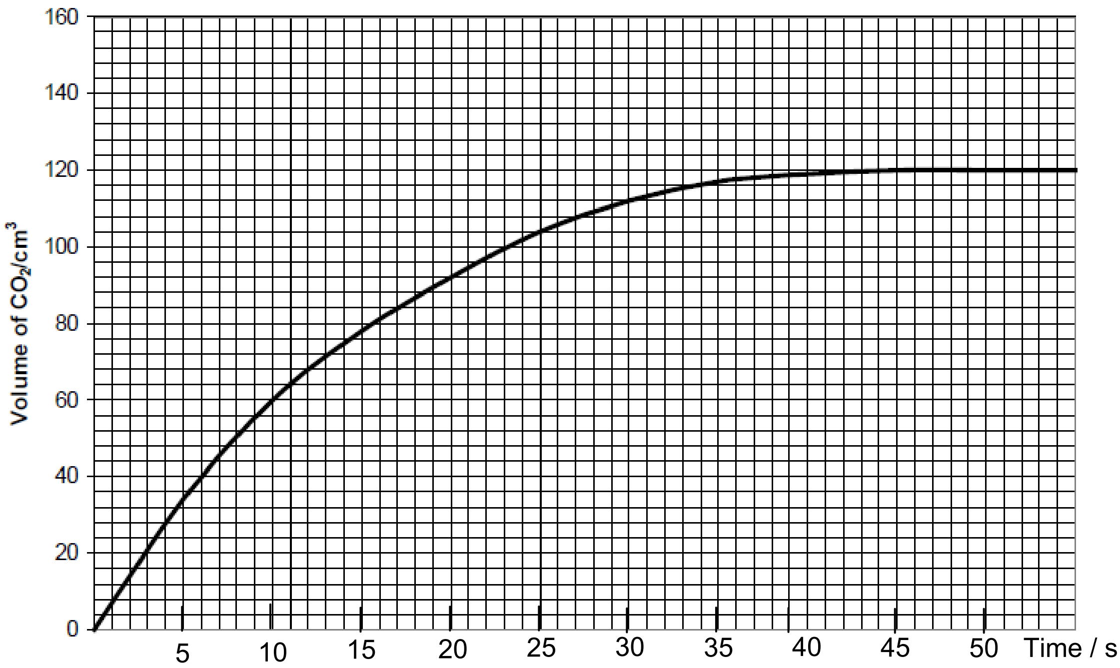
$$Na_2CO_3 + 2CH_3CH_2COOH \rightarrow 2CH_3CH_2COO^{-}Na^{+} + CO_2 + H_2O$$

Amount of HC*l* used = 0.1 x 100/1000
= 0.0100 mol

Amount of CO_2 liberated = 0.0100 / 2
= 0.00500 mol

Volume of CO_2 liberated = 0.00500 x 24000
= 120 cm^3

(ii) Plot a graph of volume of CO_2 against time for the experiment. Label this graph as Experiment **I**.



(iii) Use your graph to determine the order of reaction with respect to **R**, showing your working clearly. Hence give the rate equation for the reaction.

Since half lives are constant at about 10 seconds, it is first order reaction with respect to **R**.

Rate = k[**R**]

(iv) The experiment was repeated using 100 cm^3 of **R** with an initial concentration of 0.0500 mol dm^{-3}. On the graph you have plotted for **(b)(ii)**, sketch the graph of volume of CO_2 against time for this new experiment. Label the graph as Experiment **II**.

To show on graph:
Max volume = 60 cm^3
Half-lives = 10 s

(v) State how the rate constant and rate of reaction will be affected when the experiment is conducted at a higher temperature. [11]

Both rate constant and rate of reaction will increase.

[Total: 20]

End of Paper